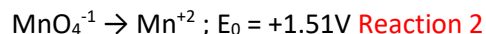
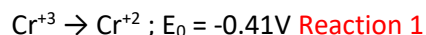


Question Set 11

- 1) Consider the two half reactions, which occur in water, and their respective standard reduction potentials, E_0 , below:



- a) Explain which half reaction should be the cathodic reaction, and which one should be the anodic reaction in order for an overall reaction to be spontaneous. Use an equation, or equations, in your answer. State the meaning of all the variables in the equation(s).

$E_0(\text{cell}) = E_{0, \text{cathode}} - E_{0, \text{anode}}$

For a reaction to be spontaneous, $E_0(\text{cell}) > 0$

Possibility 1: Reaction 1 cathode, reaction 2 anode: $E_0(\text{cell}) = -0.41 - 1.51 = -1.92\text{ V} (<0)$ NOT spontaneous

Possibility 2: Reaction 2 cathode, reaction 1 anode: $E_0(\text{cell}) = 1.51 - (-0.41) = +1.92\text{ V} (>0)$ **spontaneous**

This means reaction 2 cathode (reduction, forward direction), reaction 1 anode (oxidation, backward direction).

- b) Write a balanced overall equation for the spontaneous reaction. You must include the missing electrons, reactants and products. Assume the reaction occurs in acidic medium. Show balanced half-reactions as part of your answer.

Cathode: forward reaction 2 ($\text{MnO}_4^{-1} \rightarrow \text{Mn}^{+2}$); anode: backward reaction 1 ($\text{Cr}^{+2} \rightarrow \text{Cr}^{+3}$)

Cathode subreaction (acidic medium: add H^+ and H_2O) :



Confirm that cathode needs electrons in reactants.

Anode subreaction:



Confirm that anode needs electrons in products.

For overall cell reaction anode subreaction x 5 (for equal electrons)



OR:



- c) Write the Nernst equation. State the meaning of all the variables. Assume 25°C , therefore $2.303 \cdot \text{RT}/\text{F} = 0.0592\text{V}$.

See LN7

- d) For the overall reaction you showed in b), calculate what ratio of reactants and product concentrations (Q) you will need to construct a voltaic cell that has a starting operating potential of 2V. Show all steps in your calculations.

$$E = E_0 - \frac{0.0592V}{n} \log Q$$

Therefore:

$$2 = 1.92V - \frac{0.0592V}{5} \log Q$$

$$0.08 = -\frac{0.0592V}{5} \log Q \rightarrow -6.76 = \log Q \rightarrow Q = 10^{-6.76} \rightarrow Q = 1.74 \times 10^{-7}$$

- e) If you use double the amount of all reagents and products from those you calculated in your balanced reaction, what is the number of moles of electrons transferred in the reaction? How about for 3 times the amounts in the balanced equation?

This is a RQ: double the reagents, double the electrons. 3x reagents, 3x electrons

- f) For a situation like e) discuss how you would calculate the cell potential. Does the potential change depending on the number of moles you use? If so, how? Or how not?

This is a RQ: You could change the equation (n), but then Q ALSO changes and affects $[X]^{\text{moles}}$ for each reactant or product. As you change this, and this is inside a log, you increase/change n but then you would have to take into account a reduction/increase inside the log:

For example: $-(0.0592/n) \log Q$, if double the number of moles, double number of electrons:

This becomes: $-(0.0592/2n) \log Q^2 = -(0.0592 \cdot 2/2n) \log Q = -(0.0592/n) \log Q$ **NO CHANGE!**

SO, overall there is no change.

This is because E is an intrinsic property!

- g) Explain how it is possible to construct a voltaic cell in which half reaction 1 is both the cathodic and anodic reaction. What are these types of cells called? What is the E_0 for such a cell?

Concentration cells working outside of equilibrium (different concentrations: $Q \text{ NOT} = 1$), $E_0 = 0$

- h) For the cell in g), calculate the ratio of concentrations of reactants and product concentrations (Q) you will need to construct a voltaic cell that has a starting operating potential of 1V. Comment on whether this system would be a good candidate for such a cell.

$$E = -\frac{0.0592V}{n} \log Q, n=1$$

$$1 = -0.0592 \log Q \rightarrow -16.89 = \log Q \rightarrow 1.28 \times 10^{-17}$$

Q too small: VERY difficult to achieve this difference of concentrations: you cannot achieve large potentials with concentration cells. Only small potentials.